In essence, diffusion models are machine learning algorithms that are trained to remove noise, often Gaussian noise, from an input signal, like an image, through a series of steps. The goal is to reach a clean sample that closely resembles the original signal.

Generating high-quality image data is one of the strengths of diffusion models. However, a major drawback of these models is the slow reverse denoising process due to its sequential nature. Moreover, diffusion models operate in pixel space, which becomes incredibly large when creating high-resolution images, resulting in significant memory consumption. Hence, it is challenging to train and use these models for inference.

To overcome these challenges, latent diffusion models reduce the memory and compute complexity by applying the diffusion process over a lower dimensional latent space rather than the actual pixel space. This approach allows the model to generate compressed representations of the images, which is the main difference between standard and latent diffusion models.

There are three main components in latent diffusion.

1. An autoencoder (VAE).
2. A U-Net.
3. A text-encoder, e.g. CLIP's Text Encoder.

What is the autoencoder (VAE)?

The VAE model comprises of an encoder and a decoder, where the encoder is responsible for converting the image into a low-dimensional latent representation that serves as input to the U-Net model. In contrast, the decoder transforms the latent representation back into an image.